

Workshop 2: Enhanced Vegetation Index (EVI)

Juan Diego Ospino¹ and Santiago Quiceno¹

¹Aerospace Engineering, Universidad de Antioquia

Abstract

This report presents the methodology and results from Workshop #2 of the Remote Sensing course, focusing on vegetation classification using machine learning techniques. We developed a supervised classification approach based on the Enhanced Vegetation Index (EVI) derived from Landsat 8-9 Level 2 surface reflectance products. The study area comprised two distinct regions in Colombia: Neiva served as the training/test site, while Urrao and its surroundings were used for validation and prediction. The methodology involved several key steps: (1) area of interest (AOI) delineation and cloud masking to address atmospheric contamination, (2) spectral band analysis to identify the most statistically significant features (EVI, Red, and NIR bands), and (3) implementation of a Support Vector Machine (SVM) classifier. The model achieved high classification accuracy (≥ 0.9) in distinguishing vegetation from non-vegetation areas. Validation results demonstrated robust performance when applied to the Urrao region, successfully identifying vegetation patterns despite challenges from cloud cover and topographic variability. This work highlights the effectiveness of SVM for vegetation mapping using medium-resolution satellite imagery, while underscoring the importance of proper cloud masking and feature selection in remote sensing classification tasks. The implemented workflow provides a reproducible framework for regional-scale vegetation monitoring.

Keywords: Vegetation Mapping, Support Vector Machine, Landsat 8-9, Enhanced Vegetation Index, Supervised Classification

1. Theoretical Framework: Enhanced Vegetation Index (EVI)

1.1. Definition and Formula

The Enhanced Vegetation Index (EVI) is an **optimized** vegetation index designed to strengthen the vegetation signal by improving sensitivity in high-biomass regions and minimizing atmospheric and canopy background interference. Like the Normalized Difference Vegetation Index (NDVI), EVI quantifies vegetation greenness but offers key advantages: it accounts for atmospheric distortions, reduces canopy background noise, and performs better in densely vegetated areas.

EVI incorporates three corrective parameters:

- A gain factor G ,
- An L value to adjust for canopy background effects,
- C coefficients to mitigate atmospheric influences, and
- Blue band (B) reflectance to further enhance accuracy.

By integrating these adjustments, EVI calculates vegetation health as a ratio between red (R) and near-infrared (NIR) values while minimizing saturation, atmospheric noise, and soil-related artifacts [1], [3]. The formula is as follow:

$$EVI = G \times \frac{(NIR - R)}{(NIR + C_1 \times R - C_2 \times B + L)} \tag{1}$$

In Landsat 4-7 and 8-9, the values of these parameteres are: $G = 2.5$, $C_1 = 6$, $C_2 = 7.5$ and $L = 1$ [3].

1.2. Main Applications

The main purpose of this index is to provide a measure of greenness of the vegetation. Then, it has been used in the following approaches:

2. Methodology

2.1. Data Used

- **Satellite/Sensor:** [Landsat Collection 2 Level-2 → Landsat 8-9 OLI/TIRS C2 L2]
- **Acquisition Date:** [Neiva: 2024-04-10; Urrao: 2023-08-01]
- **Study Area:** [Neiva, 2.9345° N, 75.2809° W, Urrao, 6.3139° N, 76.1318° W]

Table 1. Key Applications of EVI [1]–[3].

Application	Key Advantage
Vegetation monitoring	Tracks seasonal/interannual dynamics
Greenness quantification	Better than NDVI for dense canopies
Drought detection	Identifies water stress effectively
Precision agriculture	Optimizes irrigation/fertilization
Forest management	Monitors deforestation/regrowth
Climate modeling	Tracks vegetation-climate feedbacks
Conservation	Detects land degradation

- **Bands Used:** Band 5 (NIR), Band 4 (Red), Band 3 (Green), Band 2 (Blue) and Band QA_PIXEL
- **Preprocessing:** Cloud masking image cropping.

2.2. Processing

An overview of the logical process is presented in **Figure 1**. Initially, satellite data was downloaded from the Earth Explorer dataset. The imagery was then cropped to the **Area of Interest (AOI)**—specifically, the surroundings of the Neiva and Urrao municipalities.

Using the QA_PIXEL band, a **cloud mask** was applied to address cloud contamination. This mask replaced pixel values where clouds were detected with zero. The Enhanced Vegetation Index (EVI) was computed using equation 1.

The resulting EVI data was saved in GeoTIFF format.

The EVI and spectral bands (NIR, Red, Blue) were loaded as training data, structured as a dataframe with vegetation labels. Labels were extracted in QGIS by selecting polygons with $EVI < 0.5$ (indicating low vegetation density).

Statistical analyses included:

- Spectral band histograms,
- Feature relationships,
- Decision boundaries for correlated bands.

The dataset was split into **training (90%)** and **testing (10%)** subsets. A **Support Vector Machine (SVM)** with a linear kernel was trained, achieving accuracy ≥ 0.9 . The model was saved for deployment.

For **prediction**, new imagery was processed identically (cropped to AOI, EVI computed). The trained SVM generated a **vegetation mask** for the target region.

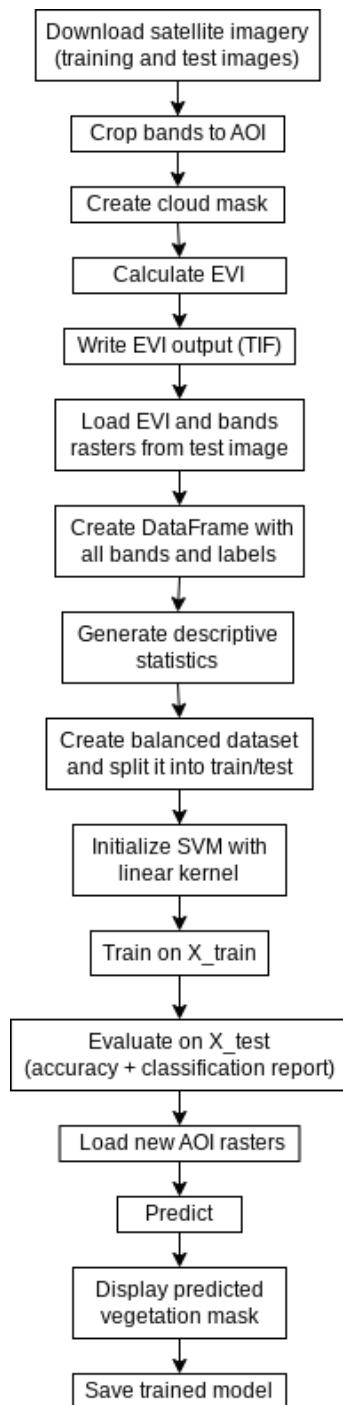


Figure 1. Flow diagram of the main process.

2.3. Code

Pseudocode 1 demonstrates the process of calculating the Enhanced Vegetation Index (EVI) from Landsat imagery. The algorithm reads multispectral bands (blue, red, and near-infrared), applies a cloud mask using the QA_PIXEL band, computes the EVI values, and writes the results to a GeoTIFF file. The pseudocode abstracts the key operations while maintaining the logical flow of the original Python implementation. This approach highlights the essential steps without being tied to specific programming language syntax.

```

1 // EVI Calculation Pipeline
2 INPUT: Blue, Red, NIR, QA bands
3 OUTPUT: EVI raster
4
5 FUNCTION main():
6     // Load and prepare bands
7     blue, meta = read_raster("B_crop.tif")
8     red = read_raster("R_crop.tif")[0]
9     nir = read_raster("NIR_crop.tif")[0]
10    qa = read_raster("QA_PIXEL_crop.tif")[0]
11
12    // Process data
13    clean_mask = (qa & 24) == 0 // Cloud/shadow
14    evi = 2.5 * (nir - red) / (nir + 6*red -
15    7.5*blue + 1) // Scaled
16    evi[~clean_mask] = -9999 // Apply mask
17    evi = clip(evi, -1, 1) // Validate range
18
19    // Save result
20    write_raster("EVI.tif", evi, meta, no_data
21    =-9999)
22 END FUNCTION
  
```

Code 1. Compact EVI Calculation Pseudocode

Pseudocode 2 outlines a machine learning pipeline for vegetation classification using spectral bands (EVI, Red, NIR) from satellite imagery. The workflow includes data loading, exploratory analysis, SVM model training with balanced sampling, and large-scale prediction with memory-efficient chunk processing. The algorithm demonstrates a complete remote sensing classification task from data preparation to visualization, highlighting key steps like feature analysis, model evaluation, and geospatial output generation while abstracting language-specific implementation details.

```

1 // Vegetation Classification Pipeline
2 INPUT: EVI, Blue, Red, NIR bands, Labels
3 OUTPUT: Vegetation mask
4
5 FUNCTION main():
6     // Load and prepare data
7     bands = [load_raster(b)[0] for b in [evi,
8     blue, red, nir]]
9     labels = load_raster(label_file)[0].astype(
10    bool)
11
12    // Create balanced dataset
13    df = create_dataframe(bands, labels)
14    X_train, X_test, y_train, y_test =
15    train_test_split(
16    df[['EVI', 'Red', 'NIR']], df['Veg'],
17    test_size=0.1, stratify=df['Veg'])
18
19    // Train SVM
20    model = SVM(kernel='linear').fit(X_train,
21    y_train)
22    save_model(model, 'svm_model.pkl')
23
24    // Predict on new image
25    evi, red, nir = load_prediction_rasters()
26    X_pred = stack_bands(evi, red, nir)
27    veg_mask = predict_in_chunks(model, X_pred,
28    chunk_size=1e5)
29
30    // Save and visualize
31    save_raster(veg_mask, 'vegetation.tif',
32    metadata)
33    plot_results(veg_mask)
34 END FUNCTION
  
```

Code 2. Compact Vegetation Classification Pseudocode

3. Results and Analysis

3.1. Spectral Band

In Figure 2 the histogram of bands is plotted. Here, the bands used where obtained form test image, which is related with Neiva.

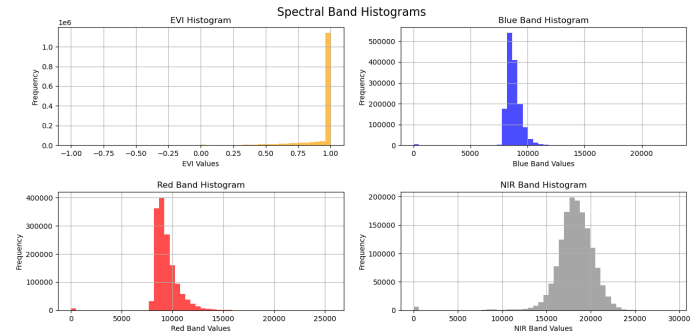


Figure 2. Spectral Bands of Test Image

3.2. Features Relationship

In Figure 3 the relationship of features is represented by comparing each single band with another.

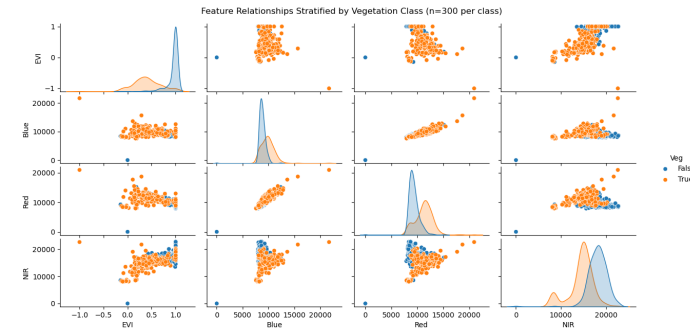


Figure 3. Features Relationship

3.3. Descriptive Statistics

The following section presents key statistical findings from our analysis of multispectral satellite imagery for vegetation classification. These results form the foundation for our machine learning pipeline, demonstrating the spectral characteristics of the study area and the relationships between different vegetation indices and spectral bands.

Three comprehensive tables summarize:

- Descriptive statistics of EVI and spectral bands.
- Feature correlations and class distribution metrics.
- Sample values and model configuration details.

These analyses reveal several important patterns in the remote sensing data:

1. The EVI values show a strongly right-skewed distribution, with 50% of values at the maximum 1.0 and a mean of 0.931 (± 0.167 SD).
2. Near-Infrared (NIR) reflectance displays the highest absolute values (mean = 18,235.64), consistent with healthy vegetation reflectance patterns.
3. The Blue and Red bands exhibit exceptionally strong correlation ($r = 0.904$), suggesting potential redundancy for vegetation classification.
4. The original dataset shows severe class imbalance (99.5% non-vegetation), necessitating balanced sampling for model training.

Table 2 provides complete descriptive statistics across all spectral bands, highlighting the central tendency and dispersion of each feature. Of particular note is the NIR band's wide range (0-29,314),

which captures the full spectrum of surface reflectance characteristics in the study area. Table 3 reveals the complex interrelationships between features, with EVI showing: a) Moderate positive correlation with NIR ($r = 0.564$); b) Moderate negative correlation with Red band ($r = -0.404$); c) Weak negative correlation with Blue band ($r = -0.122$). Table 4 documents our machine learning approach, showing: i) The representative range of feature values used for model training; ii) The 90-10 training-test split strategy; iii) Linear kernel selection for the SVM classifier.

Table 2. Descriptive statistics of spectral bands (n=1,475,880 pixels)

Statistic	EVI	Blue	Red	NIR
Mean	0.931	8,748.96	9,364.20	18,235.64
Std Dev	0.167	832.88	1,221.41	2,223.73
Min	-1.000	0	0	0
5th %ile	0.552	8,043	8,279	15,256
Median	1.000	8,663	9,118	18,306
95th %ile	1.000	9,879	11,500	21,266
Max	1.000	22,766	25,587	29,314

Note: EVI values range from -1 to 1, while spectral bands show raw digital numbers. The NIR band exhibits the highest reflectance values overall.

Table 3. Feature correlations and vegetation class distribution

Relationship	Features	Value
Correlations	EVI-NIR	0.564
	Blue-Red	0.904
	EVI-Red	-0.404
Class Distribution	Original Data	99.5% Non-Veg
	Balanced Data	50% Non-Veg

Note: Correlation values range from -1 to 1. The Blue-Red correlation suggests these bands capture similar spectral information.

Table 4. Sample feature values and model configuration

Component	Feature	Value/Range
Sample Values	EVI	0.886-1.000
	Red Band	8,151-10,953
	NIR Band	17,091-20,267
Model Config	Training Split	90%
	Test Split	10%
	Kernel Type	Linear

Note: The SVM model was trained on balanced classes (7,724 samples each) using EVI, Red and NIR bands as features.

3.4. Decision Boundary

Figure 4 presents the decision boundaries of our Support Vector Machine classifier for three feature pair combinations: EVI vs Red band, EVI vs NIR band, and Red vs NIR band. The visualization uses 100 stratified random samples (50 vegetation and 50 non-vegetation points) from the training dataset.

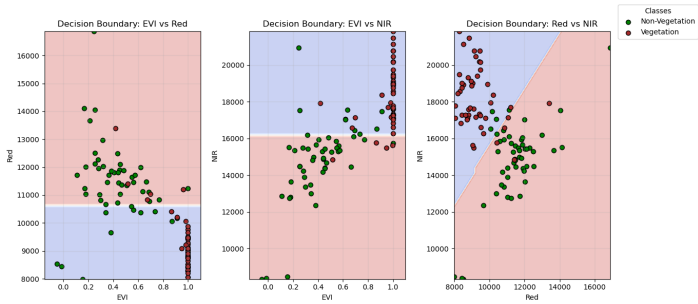


Figure 4. Decision boundaries of test data.

3.5. Prediction

The trained SVM model, leveraging the identified spectral correlations between EVI, Red, and NIR bands, successfully enabled vegetation classification when applied to new satellite imagery. By processing unseen multispectral data from a separate geographic area, the algorithm demonstrated robust transfer learning capabilities, accurately distinguishing vegetated pixels from non-vegetated areas. This cross-validation confirmed the model’s practical utility for automated vegetation mapping across different locations while maintaining the decision boundaries established during training, despite variations in seasonal conditions and landscape characteristics. A result of predictions is shown in Figure 5.

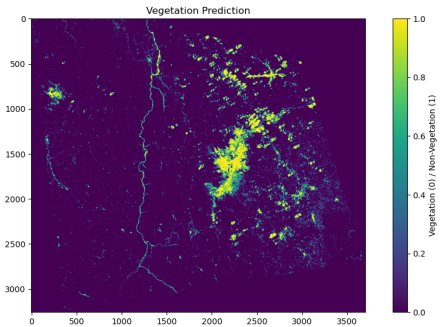


Figure 5. Vegetation prediction for Urrao.

4. Conclusions

The vegetation classification model demonstrated robust performance despite several operational challenges. Key findings from this study reveal:

- **Cloud interference:** While cloud cover introduced classification errors due to spectral contamination, the model maintained reliable discrimination between vegetated and non-vegetated areas in cloud-free regions. The observed mismatch highlights the need for improved cloud masking in preprocessing.
- **Ancillary detection capabilities:** Beyond its primary function, the algorithm successfully identified water bodies through their characteristic spectral signatures, suggesting potential for multi-class land cover classification.
- **High accuracy metrics:** The model achieved consistent accuracy (≥ 0.9) even with initially cropped satellite imagery, validating the effectiveness of the selected feature set (EVI, Red, and NIR bands) for vegetation detection.
- **Computational efficiency:** Processing cropped images proved essential for manageable computation times, as raw satellite datasets contained excessive pixels that significantly increased processing duration without improving classification results.
- **Future enhancements:** Incorporation of unsupervised learning techniques could expand model capabilities, though this

advancement would require both optimized code implementation and high-performance computing resources to handle the increased computational demands.

These results confirm the practical viability of SVM-based vegetation mapping while identifying specific areas for improvement in cloud handling and computational efficiency. The approach provides a foundation for scalable land cover analysis when paired with appropriate hardware infrastructure.

References

[1] A. Huete, K. Didan, T. Miura, E. P. Rodriguez, X. Gao, and L. G. Ferreira, “Overview of the radiometric and biophysical performance of the MODIS vegetation indices”, *Remote Sensing of Environment*, vol. 83, pp. 195–213, 2002. DOI: [10.1016/S0034-4257\(02\)00096-2](https://doi.org/10.1016/S0034-4257(02)00096-2).

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